

Dong-Joon Yi

Ph.D. Candidate, Department of Electronic Engineering, Hanyang University

Semiconductor Nanophotonics Laboratory | Advisor: Prof. Mun Seok Jeong

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Education & Careers

Hanyang University

Ph.D. candidate in Electronic Engineering (GPA 4.3 / 4.5)

- Semiconductor Nanophotonics Laboratory
- Advisor: Prof. Mun Seok Jeong

Seoul, Republic of Korea

Mar. 2024 – Aug. 2027 (est.)

Dongguk University

M.S. in Solid-state Physics (GPA 4.5 / 4.5)

- Advanced Physics on Electromaterial Laboratory
- Advisor: Prof. Kwun-Bum Chung
- Thesis: "Performance Control of InSnZnO/InGaZnO Multi-Channel-Layered Transistors with Defect Analysis of Amorphous Oxide Semiconductors"

Seoul, Republic of Korea

Sep. 2022 – Feb. 2024

Dongguk University

B.S. in Physics (GPA 4.2 / 4.5)

- Military Service in the Republic of Korea (Jun. 2019 – Apr. 2021)
- Undergraduate researcher in Advanced Physics on Electromaterial Laboratory (Jul. 2021 – Aug. 2022)

Seoul, Republic of Korea

Mar. 2016 – Aug. 2022

Scholarships & Awards

2022, 2023	Physics Department Alumni Scholarship	Dongguk University
2021	Dongguk 1.1.1 Scholarship	Dongguk University
2022, 2023	Dongguk Future Talent Scholarship	Dongguk University
2024	IC-PBL Graduate Championship, 3rd Place	Hanyang University
2025	HY-BrainKorea G3 Award, 2nd Place	Hanyang University
2026	Outstanding Presentation Award, KPS Spring Meeting, Oral Session	Korean Physical Society

Skills

Spectroscopy	Optical spectroscopy (Raman, Photoluminescence), Ellipsometry, Deep-Level Transient Spectroscopy (DLTS), Tip-Enhanced Nanospectroscopy
Machine Learning	Physics-based data analysis, Deep learning for experimental data, Explainable Artificial Intelligence (XAI)
Simulation	Finite-Difference Time-Domain (FDTD) simulations, Inverse design with adjoint optimization
Others	Electrical characterization (I-V, etc.), X-ray Photoelectron Spectroscopy (XPS), Photolithography, Sputtering

Research

My research focuses on quantitative defect spectroscopy and experimental data analysis for two-dimensional (2D) materials and oxide semiconductors. I aim to extract physically meaningful parameters from electrical and optical measurements and relate them directly to material properties and device behavior, with an emphasis on interpretability and reproducibility.

1. Defect & Optical Spectroscopy

- Quantitative analysis of intrinsic and extrinsic defects in 2D materials and oxide semiconductors.
- Extraction of defect energy levels, densities, and carrier dynamics using Deep-Level Transient Spectroscopy (DLTS) and Photo-Induced Current Transient Spectroscopy (PICTS).
- Raman / Photoluminescence (PL) spectroscopy for probing defect states, doping, strain, and phase transitions.
- X-ray Photoelectron Spectroscopy (XPS) and spectroscopic ellipsometry (SE) for chemical composition, bonding states, and optical properties.
- Tip-Enhanced Raman Spectroscopy (TERS) for spatially resolved defect and structural inhomogeneity analysis beyond the diffraction limit.

2. Explainable Artificial Intelligence (XAI) for Experimental Data

- Development of analysis pipelines for spectroscopic and electrical datasets, including preprocessing, feature extraction, modeling, and validation.

- XAI to identify physically relevant data features.
- Connection of AI-extracted features to defect-related mechanisms and material properties under data-limited experimental conditions.

3. Advanced Electronic & Photonic Devices

- Fabrication and characterization of electronic devices based on 2D materials and oxide semiconductors, including logic and neuromorphic devices.
- Quantitative analysis of device electrical responses (e.g., I-V characteristics and transient behavior) to identify dominant transport and trapping mechanisms.
- Finite-Difference Time-Domain (FDTD) simulations to evaluate optical responses and electric-field distributions in nano- to micro-scale structures, supporting the design and analysis of photonic components.

My work integrates defect-resolved spectroscopy, explainable analysis of experimental data, and device-level characterization to obtain quantitatively grounded insight into solid-state materials. The primary objective is to extract physically meaningful parameters from measurements and connect them directly to material behavior and device performance.

Publications

Google Scholar | [†] First author * Corresponding author

10. Y. Cho, J. Yoo, S. Yang, **D.-J. Yi**, S. M. Lee, M. S. Jeong, J. Choo, "Position: Significant impact of numerical precision in scientific machine learning," *International Conference on Machine Learning (ICML), Position Paper Track*, (2026), OpenReview
9. H. C. Suh[†], T. Kim[†], **D.-J. Yi**, D. H. Kim, S. H. Kim, J. Yoo, S. Bang, D. Lee, J. H. Kim, Y. S. Won, K. K. Kim, M. S. Jeong*, "Prediction and assessment of nanoprobe for tip-enhanced Raman spectroscopy: Data-driven artificial intelligence approach," *Materials & Design*, (2026), DOI:10.1016/j.matdes.2026.116066
8. T. G. Weldemhret[†], **D.-J. Yi**[†], K. Jeong, K.-B. Chung*, "Evaluation of the high mobility and stability of InGaZnO/InSnZnO bilayer thin-film transistors via quantitative defect analysis," *Surfaces and Interfaces*, (2025), DOI:10.1016/j.surf.2025.106870
7. S. Bang[†], C. Lee[†], D. Choi, D. Y. Park, D. H. Kim, D. Lee, **D.-J. Yi**, J. Song, S. J. Yun, D.-W. Kim*, M. S. Jeong*, "High Performance P-Channel Transistor Based on Amorphous Tellurium Trioxide," *Advanced Materials*, (2025), DOI:10.1002/adma.202504948
6. D. H. Kim[†], J. Yoo, H. C. Suh, Y. S. Won, S. H. Kim, **D.-J. Yi**, B. G. Jeong, C. Lee, D. Lee, K. K. Kim, S. M. Lee, E. K. Koh, M. S. Jeong*, "Anomalous phonon softening with inherent strain in wrinkled monolayer WSe₂," *Advanced Materials*, (2025), DOI:10.1002/adma.202419414
5. K. Jeong[†], D. Y. Shin, J.-M. Park, **D.-J. Yi**, H. Hong, H.-S. Kim*, K.-B. Chung*, "Noncontact Monitoring and Imaging of the Operation and Performance of Thin-Film Field-Effect Transistors," *Advanced Science*, (2025), DOI:10.1002/adv.202407923
4. A. Song[†], M. J. Kim, **D.-J. Yi**, S. Kwon, D.-W. Kim, S. Kim, J.-H. Bae, S. Park, Y. S. Rim*, K.-S. Jeong*, K.-B. Chung*, "Control of sensitivity in metal oxide electrolyte gated field-effect transistor-based glucose sensor by electronegativity modulation," *Scientific Reports*, (2024), DOI:10.1038/s41598-024-76885-x
3. H. Hong[†], M. J. Kim, **D.-J. Yi**, D. Y. Shin, Y.-K. Moon, K.-B. Chung*, "Quantitative dynamic evolution of unoccupied states in hydrogen diffused InGaZnSnO TFT under positive bias temperature stress," *ACS Applied Electronic Materials*, (2024), DOI:10.1021/acsaelm.4c01430
2. H. Hong[†], **D.-J. Yi**, Y.-K. Moon, K.-S. Son, J. H. Lim, K. Jeong, K.-B. Chung*, "Double gated a-InGaZnO TFT properties based on quantitative defect analysis and computational modeling," *IEEE Transactions on Electron Devices*, (2024), DOI:10.1109/TED.2023.3347503
1. H. Hong[†], M. J. Kim, **D.-J. Yi**, Y.-K. Moon, K.-S. Son, J. H. Lim, K. Jeong*, K.-B. Chung*, "Quantitative analysis of defect states in InGaZnO within 2 eV below the conduction band via photo-induced current transient spectroscopy," *Scientific Reports*, (2023), DOI:10.1038/s41598-023-40162-0

Patents

3. K.-B. Chung, K. Jeong, **D.-J. Yi**, (Dongguk University Industry Academy Cooperation Foundation). "Device Array Analysis Method and Analysis Apparatus Therefor." *US Patent*, 18771046.
2. K.-B. Chung, K. Jeong, **D.-J. Yi**, (Dongguk University Industry Academy Cooperation Foundation). "Analysis Method of Semiconductor Device and Analysis Apparatus for the Method." *Patent, Republic of Korea*, 10-2024-0049532.
1. K.-B. Chung, K. Jeong, **D.-J. Yi**, H. Hong, (Dongguk University Industry Academy Cooperation Foundation). "Method of Confirming Performance of Secondary Battery and Apparatus of Performing the Same." *Patent, Republic of Korea*, 10-2024-0049532.

Research Fellowship

NRF Basic Science Research Program

50 M KRW

Principal Investigator

Sep. 2025 – Aug. 2027

- Project: “Deep-Level Transient Spectroscopy and Nanospectroscopy for Ultrafine Defect Analysis Using Explainable AI in 2D Quantum Semiconductors”
- Funded by the Ministry of Education through the National Research Foundation of Korea (NRF).